Chapter 10. Linear Regression

Least squares, permutation and bootstrap inference, robust regression and M-estimation

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Overview

This chapter treats nonparametric inference in linear regression. Linear regression, which models the relation among variables to infer it or to predict the response from explanatory variables, is the most basic method. We first study parametric and nonparametric inference for the regression coefficients via least squares, the most basic estimation method; then, briefly, robust estimators usable when the regression model is right but there are outliers or the error distribution deviates badly from normality. Nonparametric regression with general regression curves (not lines) is treated in Chapter 12.

10.1 Simple linear regression

Given bivariate observations $(x_1, y_1), \ldots, (x_n, y_n)$, the simple linear regression model is the simplest model to explain and predict the response ("Y") by the explanatory variable ("X"); it assumes the relation of the two variables is "linear":

$$y_i = \alpha + \beta x_i + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.1}$$

Apart from the error ϵ_i, y_i is the linear function $\alpha + \beta x_i$ of x_i; the errors ϵ_i are uncorrelated and all follow a distribution F_ϵ with mean 0 and variance σ^2. The values $x_1, \ldots, x_n$ may be randomly sampled from some population, observed by chance, or fixed in advance by the data collector. (In the last case resampling the "X" variable may not be right.) In any case, ordinary inference in linear regression is conditional on the observed $x_1, \ldots, x_n$.

The errors of (10.1) are assumed only to have equal mean and variance; the shape of their distribution is unspecified. If the errors are assumed normal, (10.1) is a parametric model; without assumptions on $\epsilon_i \sim F_\epsilon$ it is a semiparametric model: the regression part is parametric and the error distribution nonparametric.

Example 10.1

Using World Bank data on 153 countries in 2020, we model the happiness score by the linear relation with GDP (`worldbank2020.csv`). With GDP as explanatory variable and happiness (Score) as response, the scatter plot (Figure 10.1 of the book) shows the relation is roughly explained by the simple linear model (10.1).

10.1.1 Least squares estimators

The most basic estimators of the coefficients α, β are the least squares estimators

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}.$$

The estimate of the regression line $y = \alpha + \beta x$ is $\hat{y} = \hat{\alpha} + \hat{\beta} x$, and the residuals at the observations (x_i, y_i) are $e_i = y_i - \hat{y}_i \equiv y_i - (\hat{\alpha} + \hat{\beta} x_i)$. The estimator of the error variance σ^2 is

$$\hat{\sigma}^2 = s^2 = \frac{\sum_{i=1}^n e_i^2}{n-2}.$$

Inference on the unknown coefficient β is usually based on the slope estimate $\hat{\beta}$. When the model (10.1) is true, $E(\hat{\beta}) = \beta$ and $\mathrm{Var}(\hat{\beta}) = \sigma^2 / S_{XX}$; writing only mean and variance,

$$\hat{\beta} \sim (\beta, \sigma^2 / S_{XX}). \tag{10.2}$$

Proof (supplied). $\hat\beta = \sum_i w_i y_i$ with $w_i = (x_i - \bar x)/S_{XX}$, $\sum_i w_i = 0$, $\sum_i w_i x_i = 1$, $\sum_i w_i^2 = 1/S_{XX}$. So $E\hat\beta = \sum_i w_i(\alpha + \beta x_i) = \beta$ and, the ϵ_i being uncorrelated with variance σ^2, $\operatorname{Var}\hat\beta = \sigma^2 \sum_i w_i^2 = \sigma^2/S_{XX}$. ■

To approximate the standard error $\operatorname{se}(\hat\beta) = \sigma/\sqrt{S_{XX}}$ the estimated standard error

$$\widehat{\operatorname{se}}(\hat\beta) = \frac{s}{\sqrt{S_{XX}}} \tag{10.3}$$

is used. (10.2) says nothing about the distribution of $\hat\beta$; if the errors are normal, $\hat\beta \sim N(\beta, \sigma^2/S_{XX})$ is known.

Errors and residuals. When model (10.1) is true, the errors satisfy $\epsilon_i \sim (0, \sigma^2)$, uncorrelated, but the residuals satisfy $e_i \sim (0, \sigma^2(1-h_i))$, weakly correlated, where $h_i = n^{-1} + (x_i - \bar x)^2/S_{XX}$ is the *leverage* of the ith observation.[1] When n is small or the observations are not spread evenly, the h_i can differ a lot, so the **adjusted residuals**

$$r_i = e_i/\sqrt{1-h_i}, \qquad i = 1, \dots, n,$$

are also used.

$\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ **and** $\operatorname{Var}(r_i) = \sigma^2$ **(supplied; Exercise 10.2).** With H the hat matrix, $\mathbf{e} = (I-H)\mathbf{y} = (I-H)\boldsymbol\epsilon$ since $(I-H)X = 0$. So $E\mathbf{e} = 0$ and $\operatorname{Var}(\mathbf{e}) = \sigma^2(I-H)(I-H)' = \sigma^2(I-H)$, H being a symmetric projection. In simple regression $H_{ii} = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{XX}} = h_i$, so $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$, and $Er_i = 0$, $\operatorname{Var}(r_i) = \sigma^2$. The off-diagonal entries $-\sigma^2 H_{ij}$ give the weak correlations. ■

10.1.2 Inference under normality

If model (10.1) is (approximately) true and the errors are (approximately) normal, or the sample is large enough for the CLT normal approximation, normal theory gives inference on β. Under the model with normal errors,

$$\hat\beta \sim N(\beta, \sigma^2/S_{XX}), \qquad t = \frac{\hat\beta - \beta}{\widehat{\operatorname{se}}(\hat\beta)} = \frac{\hat\beta - \beta}{s/\sqrt{S_{XX}}} \sim t_{n-2}, \tag{10.4}$$

so the $1-\alpha$ interval for β is exactly

$$\beta \in \hat\beta \pm t_{\alpha/2, n-2}\frac{s}{\sqrt{S_{XX}}}, \tag{10.5}$$

and the p-value for $H_0 : \beta = 0$ against $H_1 : \beta > 0$ is exactly $p = P(T_{n-2} \ge t_0)$, $t_0 = \frac{\hat\beta - 0}{\widehat{\operatorname{se}}(\hat\beta)}$.

Example 10.2

For the GDP–happiness regression of Example 10.1, test $H_0 : \beta = 0$. From the data $\hat\beta = 2.31$, $\widehat{\operatorname{se}}(\hat\beta) = 0.15$, $t_0 = 15.08$; this value is very large, so even without the t_{n-2} distribution ($n = 153$) a tiny p-value is expected. For the p-value to be accurate, the error distribution must not depend on the explanatory variable (GDP) and must be approximately normal. To check this, compute the adjusted residuals and visualize them by a scatter plot and histogram:

```
h = read.csv("data/worldbank2020.csv")
y = h$Score
x = h$GDP.per.capita
```

[1] The leverage h_i is large when x_i is far from the mean $\bar x$; then, like a heavy weight far from the fulcrum of a lever, the regression line changes much. Large leverage indicates a possible outlier or influential point, which needs attention in interpreting the fitted model.

```
(lm.out = lm(y ~ x))
adjusted.res = lm.out$residuals / sqrt(1 - hatvalues(lm.out))
```

The errors are roughly homoscedastic, but their distribution has a slightly longer left tail; normality is roughly satisfied.

If model (10.1) is (approximately) true but the error distribution is very different from normal, the accuracy of intervals and tests based on normal theory is not guaranteed. Alternatives are resampling-based permutation tests and the bootstrap.

10.1.3 Permutation tests for the regression coefficient

In simple regression, $H_0 : \beta = 0$ against $H_1 : \beta > 0$ for the slope can be tested by a permutation test, assuming the errors $\{\epsilon_1, \ldots, \epsilon_n\}$ are exchangeable (satisfied if they are independent with the same distribution); the test statistic can simply be $\hat{\beta}$. Without normality the null distribution of $\hat{\beta}$ cannot be obtained from (10.4).

Simple regression and its least squares estimator are closely related to correlation, in particular the sample correlation (Chapter 5). When the "X" values are chosen and fixed in advance, the population correlation ρ estimated by the sample correlation is hard to define. But whether the explanatory variable is random or fixed in model (10.1), if $H_0 : \beta = 0$ is true,

$$y_i = \alpha + \epsilon_i, \qquad i = 1, \ldots, n, \tag{10.6}$$

and the explanatory and response variables are unrelated. The errors $\{\epsilon_1, \ldots, \epsilon_n\}$ are exchangeable, so the responses $\{y_1, \ldots, y_n\}$ are too. As bivariate observations, (x_1, y_1) and (x_1, y_2) have the same distribution, so the permutation distribution of $\hat{\beta}$ can be generated by fixing the "X" values and shuffling only the "Y" values. (The same shuffles were used in the permutation test of correlation.) There are $n!$ shuffles, very many, so a random permutation test is recommended:

1. Compute the least squares estimate $\hat{\beta}_{\text{obs}}$ of β in model (10.1) from $\{(x_1, y_1), \ldots, (x_n, y_n)\}$.
2. Randomly shuffle the order of the responses to get a shuffled sample $\{(x_1, y_1^*), \ldots, (x_n, y_n^*)\}$, where $(y_1^*, \ldots, y_n^*)$ is drawn by sequential simple sampling without replacement from $\{y_1, \ldots, y_n\}$.
3. Compute $\hat{\beta}^*$ from the shuffled sample.
4. Repeat steps 2–3 $M = 1999$ times to generate the permutation distribution of $\hat{\beta}$ as the empirical distribution of $\{\hat{\beta}^{*(0)}, \ldots, \hat{\beta}^{*(M)}\}$ ($\hat{\beta}^{*(0)} := \hat{\beta}_{\text{obs}}$).
5. For the upper-tailed $H_1 : \beta > 0$, $p \approx \sum_{i=1}^{M} I(\hat{\beta}^{*(i)} \geq \hat{\beta}_{\text{obs}})/(M+1)$.

For lower-tailed or two-sided tests change the definition of extreme. $M = 1999$ is usual (more for accuracy). Since exchangeability of the responses under the null was used, this permutation test is exact conditionally on the observations.

Above the statistic was $\hat{\beta}^* = S_{XY^*}/S_{XX}$ ("$*$" = value after shuffling). This permutation test coincides with permutation tests using any of:

- $S_{XY^*} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i^* - \bar{y})$ or $\sum_{i=1}^{n} x_i y_i^*$;
- $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$, where $\widehat{\text{se}}(\hat{\beta}^*) = [\sum_{i=1}^{n}(e_i^*)^2/\{(n-2)S_{XX}\}]^{1/2}$, $e_i^* = y_i^* - \hat{\alpha}^* - \hat{\beta}^* x_i$.

As in two-sample permutation tests (sum of one sample, difference of means, two-sample t all giving the same test), S_{XY^*} and t^* are monotone increasing functions of $\hat{\beta}^*$, so any of them gives the same test.

Why t^* is increasing in $\hat{\beta}^*$ (supplied; Exercise 10.3). S_{XX} and $S_{YY} = \sum(y_i^* - \bar{y})^2$ are the same for every shuffle, and $\sum(e_i^*)^2 = S_{YY} - (\hat{\beta}^*)^2 S_{XX}$, so

$$\widehat{\mathrm{se}}^2(\hat{\beta}^*) = \frac{S_{YY} - (\hat{\beta}^*)^2 S_{XX}}{(n-2)S_{XX}}, \qquad t^* = \frac{\hat{\beta}^* \sqrt{(n-2)S_{XX}}}{\sqrt{S_{YY} - (\hat{\beta}^*)^2 S_{XX}}},$$

which is strictly increasing in $\hat{\beta}^*$ on $(\hat{\beta}^*)^2 S_{XX} < S_{YY}$. Hence $\{\hat{\beta}^* \geq \hat{\beta}\} = \{t^* \geq t\}$ for every shuffle. ■

Example 10.3

For the happiness–GDP regression, the permutation test of $H_0 : \beta = 0$ against $H_1 : \beta > 0$ with $\hat{\beta}$ as statistic. The random permutation test below gave $p = 1/2000$; compared with the observed $\hat{\beta}_{\text{obs}} = 2.31$, the permutation distribution under the null almost never produces such a value, so the linear relation between the variables is very significant.

```
> n = length(x)
> Sxx = sum( (x - mean(x))^2 )
> Sxy = sum( (x - mean(x)) * (y - mean(y)) )
> beta.hat = Sxy / Sxx
>
> M = 1999
> beta.hat.perm = replicate(M, {
+   yperm = sample(y,n, replace = FALSE)
+   sum( (x - mean(x)) * (yperm - mean(yperm))) / Sxx
+ })
> (p = mean( c(beta.hat, beta.hat.perm) >= beta.hat ))
[1] 5e-04
```

10.1.4 Bootstrap confidence intervals for the regression coefficient

Permutation tests are exact, but they are hard to use for confidence intervals because of the computation. For nonparametric interval estimation of β the bootstrap is used. Chapter 7 gave two bootstraps of bivariate data; given $(x_1, y_1), \ldots, (x_n, y_n)$ and the linear model $y_i = \alpha + \beta x_i + \epsilon_i$:

- **Case (pairs) bootstrap:** resample the n pairs $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ by simple sampling with replacement. Unlike shuffling in permutation tests, which destroys the correlation between "X" and "Y", the bootstrap always draws the pairs (x_i, y_i) together, preserving the linear relation.
- **Residual bootstrap:** assuming the linear model is true, the errors $\epsilon_i \sim F_\epsilon$ in $y_i = \alpha + \beta x_i + \epsilon_i$ carry the randomness of sampling from the population. To approximate F_ϵ from the data, use the residuals $\{e_1, \ldots, e_n\}$ computed from $\hat{\alpha}, \hat{\beta}$ and resample them with replacement. Since the residual variances $\sigma^2(1 - h_i)$ differ slightly with the leverage, resampling the adjusted residuals $\{r_1, \ldots, r_n\}$ is preferred. With bootstrap residuals $r_1^*, \ldots, r_n^*$, the bootstrap sample is

$$(x_i^*, y_i^*) = (x_i, \hat{\alpha} + \hat{\beta} x_i + r_i^*). \tag{10.7}$$

  In the residual bootstrap the explanatory values are not resampled: each bootstrap sample has the same "X" values as the original.

Whichever resampling is used, the bootstrap replicates of $\hat{\beta}$ are obtained by:

1. Get a bootstrap sample $D^* = \{(x_1^*, y_1^*), \ldots, (x_n^*, y_n^*)\}$ by the case or residual bootstrap.

2. From D^* compute $\hat\beta^* = S_{X^*Y^*}/S_{X^*X^*}$ and $t^* = (\hat\beta^* - \hat\beta)/\widehat{\mathrm{se}}(\hat\beta^*)$, $\widehat{\mathrm{se}}(\hat\beta^*) = [\sum_{i=1}^n (e_i^*)^2/\{(n-2)S_{X^*X^*}\}]^{1/2}$.
3. Repeat steps 1–2 B times to get $\{(\hat\beta^{*b}, \widehat{\mathrm{se}}(\hat\beta^{*b}), t^{*b}) : b = 1, \ldots, B\}$.

(For the residual bootstrap, $S_{X^*X^*} = S_{XX}$, which saves computation.) The empirical distribution of $\{\hat\beta^{*b}\}$ is the bootstrap sampling distribution of $\hat\beta$, giving bootstrap estimates of bias and standard error. In linear regression $\hat\beta$ is unbiased and its standard error is (10.3), so these are not really needed; but when normal errors are hard to assume or the sample is small, bootstrap intervals are more accurate than normal-theory ones.

From the replicates one gets the basic, bootstrap-t and (adjusted) percentile intervals of Chapter 8. For regression coefficients, of the two second-order accurate intervals (bootstrap-t and BCa), the bootstrap-t is recommended. It is not used often in general because the standard error needs a second-level bootstrap; in regression the standard error of $\hat\beta$ has a formula, so no second level is needed. That is why $\widehat{\mathrm{se}}(\hat\beta^*)$ and t^* were computed above along with $\hat\beta^*$.

Example 10.4

The 95% bootstrap-t interval for β in the happiness–GDP regression. With $Q_H^*(\alpha/2)$, $Q_H^*(1-\alpha/2)$ the 2.5% and 97.5% quantiles of the replicates $t^{*1}, \ldots, t^{*B}$, the interval is

$$\beta \in (\hat\beta - \widehat{\mathrm{se}}(\hat\beta) Q_H^*(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta) Q_H^*(\tfrac{\alpha}{2})). \tag{10.8}$$

The code records only t^{*b} from $B = 1999$ case-bootstrap resamples; computing t^* is lengthy, so a function `t.pivot()` is defined.

```
t.pivot = function(x,y, beta0){
  n = length(x)
  Sxx = sum( (x - mean(x))^2 )
  Sxy = sum( (x - mean(x)) * (y - mean(y)) )
  beta.hat = Sxy / Sxx
  res = y - mean(y) - beta.hat * ( x - mean(x))
  se.beta.hat = sqrt( sum( res^2 ) / (n - 2) ) / sqrt(Sxx)
  t = (beta.hat - beta0)  / se.beta.hat
  t
}
set.seed(1)
B = 1999
boot.t = replicate(B, {
    boot.id = sample(1:n,n, replace = TRUE) # case resampling
    xboot = x[boot.id]
    yboot = y[boot.id]
    t.pivot(xboot, yboot, beta.hat)
})
res = y - mean(y) - beta.hat * ( x - mean(x))
s = sqrt( sum( res^2 ) / (n - 2) )
se.beta.hat = s / sqrt(Sxx)
boot.t.ci = beta.hat - se.beta.hat * quantile(boot.t, probs = c
    (1-0.025, 0.025))
```

Compared with the normal-theory interval (10.5):

```
> boot.t.ci
   97.5%     2.5%
2.024506 2.597004
> (norm.appr.ci = beta.hat + se.beta.hat * qt(c(0.025, 1-0.025),
    df = n-2))
[1] 2.012202 2.618830
```

With the residual bootstrap the coding is more efficient (S_{XX} is common to all bootstrap samples). Ignoring that, the residual-bootstrap interval, continuing Example 10.2:

```
> set.seed(1)
> boot.t.res = replicate(B, {
+   boot.res = sample(adjusted.res,n, replace = TRUE) # residual
    resampling
+   yboot = lm.out$fitted.values + boot.res
+   t.pivot(x,yboot, beta.hat)
+ })
> (res.boot.t.ci = beta.hat - se.beta.hat * quantile(boot.t.res,
    probs = c(1-0.025, 0.025)))
   97.5%     2.5%
2.024342 2.611618
```

The residual-bootstrap interval $\beta \in (2.02, 2.61)$ is almost the same as the normal-theory interval.

The case and residual bootstraps differ, and the residual-bootstrap interval is closer to the normal-theory one. Recall the derivation of (10.5): the explanatory variable is treated as fixed, and the interval is valid when the linear model is true; i.e. it is inference conditional on the observed explanatory values. The residual bootstrap also fixes the explanatory values and resamples, so it gives results closer to the usual normal theory. In Example 10.1 the errors are approximately normal, so the intervals hardly differ; when errors are far from normal the bootstrap intervals are more accurate.

The case bootstrap allows accurate estimation of the parameter β even if the linear model is wrong (the relation is not linear). Then β is no longer a coefficient of a linear model but can be understood as the value minimizing the mean squared error $E(Y-\alpha-\beta X)^2$. The case bootstrap gives inference robust to model misspecification, but when the model is true it is less efficient than the residual bootstrap. In bivariate data from a designed experiment, the case bootstrap changes the explanatory values (the design information that should be fixed) in every resample, which can also be a drawback (Davison and Hinkley, 1997).

10.1.5 Bootstrap tests for the regression coefficient

Chapter 9 tested parameters by bootstrapping their estimates; tests of regression coefficients proceed the same way. Assuming model (10.1) and testing $H_0 : \beta = 0$ against $H_1 : \beta > 0$, the statistic is

$$t_0 = \frac{\hat{\beta} - 0}{\widehat{\text{se}}(\hat{\beta})}, \tag{10.9}$$

rejecting H_0 when t_0 is large. Bootstrap resampling for the test can be resampling under the null or resampling with a pivot.

Nonparametric bootstrap resampling under the null. Bootstrap samples are drawn assuming the null is true. If $\beta = 0$,

$$y_i = \alpha + \epsilon_i, \qquad i = 1, \ldots, n, \tag{10.10}$$

and x and y are unrelated (independent). Two bootstrap resamplings are possible:

1. **Case bootstrap.** Draw $x_1^*, \ldots, x_n^*$ by simple sampling with replacement from $\{x_1, \ldots, x_n\}$ and, independently, $y_1^*, \ldots, y_n^*$ from $\{y_1, \ldots, y_n\}$. From the bootstrap sample compute $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$; the empirical distribution of $\{t^{*1}, \ldots, t^{*B}\}$ over B repetitions is the null distribution of t_0 (10.9). (Under the null the two variables are independent, so the distribution of (X, Y)

is the product of the distribution F_X of X and $F_Y(y) = F_\epsilon(y-\alpha)$ of Y; it is approximated by the replicates.)

2. **Residual bootstrap.** Used when the explanatory values are fixed or inference is conditional on them. The null model (10.10) still has the unknown α, so plug in its null estimate $\hat{\alpha}^0 = \bar{y}$ to get residuals
$$e_i^0 = y_i - \bar{y}, \qquad i = 1, \ldots, n, \tag{10.11}$$
and adjusted residuals $r_i^0 = e_i^0/\sqrt{1-\frac{1}{n}}$ (superscript "0" = value in the null model). Resample them with replacement as $r_1^{0*}, \ldots, r_n^{0*}$, form $(x_i^*, y_i^*) = (x_i, \hat{\alpha} + r_i^{0*})$, and compute $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$. The empirical distribution of $\{t^{*1}, \ldots, t^{*B}\}$ is the null distribution of t_0 (10.9).

Example 10.5

Testing whether $\beta > 0$ in the happiness–GDP regression, p-values by the case and residual bootstraps under the null; `t.pivot()` of Example 10.4 is reused.

```
# 1. case bootstrap, resampling under H0
set.seed(1)
t0 = t.pivot(x,y,0)
B = 1999
boot.t = replicate(B, {
  xboot = sample(x,n, replace = TRUE)
  yboot = sample(y,n, replace = TRUE)
  t.pivot(xboot,yboot, 0)
})
p.null.case = mean(c(t0,boot.t ) >= t0)
```

The statistic $t_0 = 15.1$ is very large and no replicate in `boot.t` exceeds it (Figure 10.2 of the book), so $p = 1/2000$. The residual bootstrap under the null gives $p = 1/2000$ as well:

```
# 2. residual bootstrap, resampling under H0
adjusted.res = (y - mean(y))/ sqrt( 1- 1/n)
boot.t = replicate(B, {
  boot.res = sample(adjusted.res, n, replace = TRUE)
  yboot = mean(y) + boot.res
  t.pivot(x,yboot, 0)
})
p.null.res = mean(c(t0,boot.t ) >= t0)
```

Bootstrap resampling with a pivot. If the general form of t_0,
$$t = (\hat{\beta} - \beta)/\widehat{\text{se}}(\hat{\beta}), \tag{10.12}$$
is assumed a pivot, the distribution of t_0 under $\beta = \beta_0 = 0$ (the null distribution) equals that of the pivot t. It is approximated by the bootstrap distribution of
$$t^* = (\hat{\beta}^* - \hat{\beta})/\widehat{\text{se}}(\hat{\beta}^*) \tag{10.13}$$
and used as the null distribution of t_0 (10.9). Resampling is as for bootstrap intervals: case bootstrap (resampling the pairs) or residual bootstrap (resampling the linear-model residuals $e_i = y_i - \hat{\alpha} - \hat{\beta}x_i$). Pivot resampling is the same as inverting the bootstrap-t interval.

Example 10.6

Testing whether $\beta > 0$ in the happiness–GDP regression with the pivot and the case bootstrap:

```
# 3. pivot, case bootstrap
boot.t = replicate(B, {
  boot.id = sample(1:n,n, replace = TRUE)
  xboot = x[boot.id]
  yboot = y[boot.id]
  t.pivot(xboot,yboot, beta.hat)
})
p.pivot.case = mean(c(t0,boot.t ) >= t0)
```

The replicates `boot.t` are exactly those of the bootstrap-t interval with the case bootstrap. The residual bootstrap with the pivot follows the same resampling as the residual-bootstrap interval:

```
# 4. pivot, residual bootstrap
boot.t = replicate(B, {
boot.res = sample(adjusted.res,n, replace = TRUE)
  yboot = lm.out$fitted.values + boot.res
  t.pivot(x,yboot, beta.hat)
})
p.pivot.res = mean(c(t0,boot.t ) >= t0)
```

The four bootstrap null distributions of Examples 10.5 and 10.6 are shown as histograms in Figure 10.2 of the book; all are similar and close to the t_{151} (or standard normal) distribution.

10.2 Multiple linear regression

The multiple linear regression model explains a response Y by a linear combination of several explanatory variables $X_1, \ldots, X_p$. With n observations,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.14}$$

The errors ϵ_i are uncorrelated with mean 0 and variance σ^2, distribution F_ϵ. Regression analysis estimates the coefficients β_i $(i = 0, \ldots, p)$, gives their confidence intervals, and tests various hypotheses about them. As in simple regression, the explanatory values are assumed fixed and inference is conditional. When F_ϵ is not $N(0, \sigma^2)$, resampling-based inference can be considered, and these methods extend the resampling methods of simple regression naturally.

Example 10.7

Using World Bank 2020 data for 153 countries (`worldbank2020.csv`), fit a linear model of the happiness score (Score) on six explanatory variables: (1) GDP, (2) Social (social support), (3) Life (healthy life expectancy), (4) Freedom, (5) Generosity, (6) Corruption, social-health indices. The scatter plot matrix with sample correlations (Figure 10.3 of the book) gives a quick view. With $X_1, \ldots, X_p$ $(p = 6)$ in this order, we want intervals and significance for each coefficient. The scatter plots suggest the last two variables (Generosity, Corruption) have weak linear association with happiness. Should we use the model with all variables or with only the first four?

10.2.1 Least squares estimators

Least squares is again basic. Stacking (10.14) over $i = 1, \ldots, n$,

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}, \qquad \text{or} \qquad \mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{10.15}$$

The matrix X is the design (model) matrix; the parameters are the $p+1$ coefficients $\boldsymbol{\beta}$ and σ^2. The least squares estimator is

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y},$$

the fitted values $\hat{\mathbf{y}} = X\hat{\boldsymbol{\beta}} = X(X'X)^{-1}X'\mathbf{y}$, and $H = X(X'X)^{-1}X'$ $(n \times n)$ is the hat matrix ($\hat{\mathbf{y}} = H\mathbf{y}$). With I_n the identity, the residual vector is $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (I_n - H)\mathbf{y}$, and

$$\hat{\sigma}^2 = s^2 = \frac{\sum_{i=1}^n e_i^2}{n-p-1} = \frac{\mathbf{e}'\mathbf{e}}{n-p-1} = \frac{\mathbf{y}'(I_n - H)\mathbf{y}}{n-p-1}. \tag{10.16}$$

When model (10.14) is true,

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(X'X)^{-1}, \tag{10.17}$$

and $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = s^2(X'X)^{-1}$. If the errors are normal, $\hat{\boldsymbol{\beta}}$ is multivariate normal with mean and covariance (10.17).

Proof of (10.17) (supplied). $\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'(X\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (X'X)^{-1}X'\boldsymbol{\epsilon}$, so $E\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}$ and $\mathrm{Var}\,\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$. Also $E(\mathbf{e}'\mathbf{e}) = E\,\boldsymbol{\epsilon}'(I-H)\boldsymbol{\epsilon} = \sigma^2\mathrm{tr}(I-H) = \sigma^2(n-p-1)$, so s^2 is unbiased. ■

Errors and residuals. As in simple regression, when (10.14) is true the errors $\epsilon_i \sim (0, \sigma^2)$ are uncorrelated with the same distribution, but the residuals are weakly correlated with $e_i \sim (0, \sigma^2(1-h_i))$, with different variances (h_i is the ith diagonal element of H). So the adjusted residuals

$$r_i = e_i/\sqrt{1-h_i}, \qquad i = 1, \ldots, n, \tag{10.18}$$

are also used.

10.2.2 Inference on the coefficients under normality

The coefficient β_j and its estimate $\hat{\beta}_j$ do not simply reflect the linear relation between the jth explanatory variable and the response; they are the coefficient explaining linearly by the jth explanatory variable the variation of the response not explained by the other explanatory variables. So inference on β_j is not about the relation between the jth variable and the response but about the role of the jth variable when the multiple regression model is true. (This is why older papers and textbooks call the β_j partial regression coefficients.) Under normality, intervals and tests for β_j are exact.

With normal errors, $\hat{\boldsymbol{\beta}}$ is multivariate normal with mean and covariance (10.17). The estimate $\hat{\beta}_j$ has the marginal distribution

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2(X'X)^{-1}_{jj}), \tag{10.19}$$

where $(X'X)^{-1}_{jj}$ is the $(j+1)$th diagonal element of the $(p+1)\times(p+1)$ matrix $(X'X)^{-1}$. With $\mathrm{se}(\hat{\beta}_j) = \{\sigma^2(X'X)^{-1}_{jj}\}^{1/2}$ and its estimate $\widehat{\mathrm{se}}(\hat{\beta}_j) = \{s^2(X'X)^{-1}_{jj}\}^{1/2}$, the studentized pivot satisfies

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\widehat{\mathrm{se}}(\hat{\beta}_j)} \sim t_{n-p-1}. \tag{10.20}$$

So the $1-\alpha$ interval for β_j is exactly

$$\beta_j \in \hat{\beta}_j \pm t_{\alpha/2,n-p-1}\widehat{\text{se}}(\hat{\beta}_j), \tag{10.21}$$

and for $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ the observed $t_{j(0)} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j)$ gives exactly $p = P(|T_{n-p-1}| \geq |t_{j(0)}|)$.

Example 10.8

Fit the multiple regression with all $p = 6$ variables of Example 10.7:

```
h = read.csv("data/worldbank2020.csv")
dat = h %>% dplyr::select(4:9,3)
n = nrow(dat); p = 6
lm.out = lm(Score ~ . , data = dat)
```

The design matrix X is obtained by `model.matrix()`; its first column is all 1 and it has $p+1$ columns.

```
> y = dat$Score
> X = model.matrix(lm.out)
> head(X)
  (Intercept)   GDP Social  Life Freedom Generosity Corruption
1           1 1.285  1.500 0.961   0.662      0.160      0.478
2           1 1.327  1.503 0.979   0.665      0.243      0.495
3           1 1.391  1.472 1.041   0.629      0.269      0.408
```

The estimated standard errors of the $\hat{\beta}_j$:

```
> s = sqrt( sum ( (lm.out$residuals)^2 ) / (n - p - 1) )
> (se.hat = s *  ( solve(t(X) %*% X) %>% diag() %>% sqrt() ) )
(Intercept)         GDP      Social        Life     Freedom
  0.2272244   0.2648616   0.2799800   0.3604369   0.4151976
 Generosity  Corruption
  0.5095199   0.4878042
```

For Generosity, the estimate is $\hat{\beta}_j = 0.6203$; its 95% interval contains 0, so $H_0 : \beta_j = 0$ is not rejected.

```
> beta.hat = solve(t(X) %*% X) %*% t(X) %*% y
> beta.hat[6] + se.hat[6] * qt(1-0.025, n-p-1) * c(-1,1)
[1] -0.3867092  1.6272659
```

10.2.3 Comparing two linear models under normality

Beyond inference on each coefficient, multiple regression allows comparing models. E.g. in Example 10.7, choose between the reduced model with only the $k = 4$ explanatory variables (dropping Generosity and Corruption) and the full model with all $p = 6$. With the full model (10.14), the reduced model with the first k variables is

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, \ldots, n) \\ &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + 0 \cdot x_{i,k+1} + \cdots + 0 \cdot x_{ip} + \epsilon_i, \end{aligned}$$

the special case $\beta_{k+1} = \cdots = \beta_p = 0$ of the full model.[2] So model selection becomes the test

$$H_0 : \beta_{k+1} = \cdots = \beta_p = 0 \quad \text{vs} \quad H_1 : \text{at least one of } \beta_{k+1}, \ldots, \beta_p \text{ is not } 0. \tag{10.22}$$

[2] More general hypotheses on coefficients, e.g. $H_0 : C\boldsymbol{\beta} = 0$, also have normal-theory tests, and in such general cases the null distribution can be approximated by normal theory, permutation or bootstrap. The principle and process are the same as comparing reduced and full models, so they are not discussed here.

In testing (10.22) or comparing two models, note that the variables $X_1, \ldots, X_k$ not being tested are always in the model. The alternative is not simply "$X_{k+1}, \ldots, X_p$ are related to Y" but

H_1 : "a linear combination of $X_{k+1}, \ldots, X_p$ explains variation of Y not explained by $X_1, \ldots, X_k$",

and the null is

H_0 : "the variation of Y explained by $X_{k+1}, \ldots, X_p$ is already explained by $X_1, \ldots, X_k$".

Write the reduced (null) model as

$$\mathbf{y} = X_0\boldsymbol{\beta}_0 + \boldsymbol{\epsilon} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \tag{10.23}$$

(subscript "0" = null model). Then the full model (10.15) is

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon} = X_0\boldsymbol{\beta}_0 + X_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon}, \qquad X_1\boldsymbol{\beta}_1 = \begin{pmatrix} x_{1,k+1} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n,k+1} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_{k+1} \\ \vdots \\ \beta_p \end{pmatrix}, \tag{10.24}$$

with $q = p - k$ added (tested) variables and $X = [X_0, X_1]$. Let $H = X(X'X)^{-1}X'$ and $H_0 = X_0(X_0'X_0)^{-1}X_0'$. In the reduced model (10.23),

$$\hat{\boldsymbol{\beta}}_{0(0)} = (X_0'X_0)^{-1}X_0'\mathbf{y}, \qquad \mathbf{e}_0 = \mathbf{y} - X_0\hat{\boldsymbol{\beta}}_{0(0)} = (I_n - H_0)\mathbf{y}.$$

In the full model (10.24),

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y} = \begin{pmatrix} \hat{\boldsymbol{\beta}}_0 \\ \hat{\boldsymbol{\beta}}_1 \end{pmatrix} = \begin{pmatrix} \hat{\boldsymbol{\beta}}_0 \\ \{X_1'(I_n - H_0)X_1\}^{-1}X_1'(I_n - H_0)\mathbf{y} \end{pmatrix}$$

(in general $\hat{\boldsymbol{\beta}}_0 \neq \hat{\boldsymbol{\beta}}_{0(0)}$), with residuals $\mathbf{e} = (I_n - H)\mathbf{y}$. Comparing the residual sizes tests whether $\boldsymbol{\beta}_1$ is significantly nonzero. The reduced-model residual sum of squares is

$$SSE_0 = \sum_{i=1}^{n} e_{i(0)}^2 = \mathbf{e}_0'\mathbf{e}_0 = \mathbf{y}'(I_n - H_0)\mathbf{y},$$

the full-model one $SSE = \mathbf{e}'\mathbf{e} = \mathbf{y}'(I_n - H)\mathbf{y}$, and the gain from additionally estimating $\boldsymbol{\beta}_1$ given $\boldsymbol{\beta}_0$ is

$$SSR_{\beta_1|\beta_0} = SSE_0 - SSE = \mathbf{y}'(I_n - H_0)\mathbf{y} - \mathbf{y}'(I_n - H)\mathbf{y} = \mathbf{y}'(H - H_0)\mathbf{y}.$$

Standardizing by the full-model residual sum of squares, each divided by its degrees of freedom, the statistic for (10.22) is

$$F = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)}. \tag{10.25}$$

When the null is true (and errors normal), $F \sim F_{q,n-p-1}$, the F-statistic. With observed F_{obs}, $p = P(F_{q,n-p-1} \geq F_{\text{obs}})$.

Why $F \sim F_{q,n-p-1}$ (supplied). H and H_0 are orthogonal projections with $\text{col}(X_0) \subset \text{col}(X)$, so $HH_0 = H_0$, $H - H_0$ is a projection of rank q, and $(H - H_0)(I - H) = 0$. Under H_0, $\mathbf{y} = X_0\boldsymbol{\beta}_0 + \boldsymbol{\epsilon}$ and $(H - H_0)X_0 = 0$, $(I - H)X_0 = 0$, so $SSR_{\beta_1|\beta_0} = \boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $SSE = \boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$. For $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I)$ these are independent $\sigma^2\chi_q^2$ and $\sigma^2\chi_{n-p-1}^2$ (Cochran), so $F \sim F_{q,n-p-1}$. ■

Example 10.9

Continuing Example 10.8, test $H_0 : \beta_5 = \beta_6 = 0$ (the coefficients of Generosity and Corruption) under normality. The reduced design X_0 is the first 5 columns of X, and X_1 the remaining two.

```
> X0 = X[,1:5]
> X1 = X[,6:7]
> q = 2
```

The observed F (10.25) (`drop()` turns the 1×1 matrix into a scalar):

```
> Hat = X %*% solve(t(X) %*% X) %*% t(X)
> Hat0 = X0 %*% solve(t(X0) %*% X0) %*% t(X0)
> SSR = t(y) %*% (Hat - Hat0) %*% y
> SSE = sum(y^2) - t(y) %*% Hat %*% y
> (F.obs = drop(SSR/SSE * (n-p-1) / q))
[1] 3.599035
```

Compared with the null distribution $F_{2,153-6-1}$:

```
> (p.theory = 1- pf(F.obs, q, n-p-1))
[1] 0.02980354
```

Remark

When the reduced model omits a single variable of the full model, $q = 1$ and the F-statistic (10.25) is the square of the t-statistic $t_{j(0)}$ (10.20). Conversely, in a test of β_j, the null model $H_0 : \beta_j = 0$ is the reduced model with the $k = p - 1$ explanatory variables other than the jth.

10.3 Nonparametric inference in multiple regression

Multiple regression involves two kinds of inference: intervals and tests for one coefficient β_j, and tests of $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$ for several coefficients. When normal errors cannot be assumed even approximately, permutation tests and bootstrap intervals and tests replace normal-theory t- and F-tests.

10.3.1 Permutation tests

For the general hypothesis (10.22) by permutation, the null distribution is obtained by shuffling for the statistic F (10.25). If the number of coefficients tested is $q = p - k = 1$, the test is about that coefficient and the t-statistic t_j (10.20) can be used. For a two-sided test of one coefficient the permutation tests with F and with t_j agree, but the one-sided $H_1 : \beta_j > 0$ needs the t-statistic.

Permutation tests in regression require exchangeable errors ϵ_i. In simple regression, under the null the responses $y_1, \ldots, y_n$ were exchangeable and we shuffled them. In multiple regression under $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, \ldots, n), \tag{10.26}$$

the errors ϵ_i are exchangeable but, unfortunately, the observations y_i are not.[3] So the advantage of permutation tests, "(conditionally) exact inference," is lost. The exchangeable errors cannot be

[3] To see this simply, let $k = 1$, $\beta_1 = 1$: then $E(y_1) = \beta_0 + x_{11} \neq \beta_0 + x_{21} = E(y_2)$, so the means of y_1, y_2 differ with the explanatory values and they are not exchangeable.

shuffled (they are unobservable), and shuffling the non-exchangeable observations is unsatisfactory, so various permutation tests have been proposed. None is exact, but they are asymptotically exact as n grows, and for very large n they agree. They were developed after Efron's bootstrap and resemble bootstrap tests. With p explanatory variables, write the ith observation $(x_{i1}, \ldots, x_{ip}, y_i) = (\mathbf{x}_i, y_i)$ and the full-model data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$. We look at (1) shuffling the observations, (2) shuffling the residuals under the null, (3) shuffling the residuals of the full model.

Shuffling the observations. The simplest: treat the (non-exchangeable) responses $\{y_1, \ldots, y_n\}$ as exchangeable and shuffle them to generate the permutation distribution of F (Manly, 2006). The shuffled sample is $\mathcal{D}^* = \{(\mathbf{x}_1, y_1^*), \ldots, (\mathbf{x}_n, y_n^*)\}$, with $(y_1^*, \ldots, y_n^*)$ drawn by sequential simple sampling without replacement from $\{y_1, \ldots, y_n\}$. The random permutation test:

1. Compute F_{obs} from $\mathcal{D}$.
2. For $i = 1, \ldots, M$ $(M = 1999)$ generate a randomly shuffled sample $\mathcal{D}^{*i}$ and compute F^{*i}.
3. The p-value for (10.22) is $p \approx \sum_{i=0}^{M} I(F^{*i} \geq F_{\text{obs}})/(M+1)$ $(F^{*0} := F_{\text{obs}})$.

Shuffling the residuals under the null. Shuffle the residuals $e_{i(0)}$ (or adjusted residuals $r_{i(0)}$) instead of the exchangeable but unobservable errors (Freedman and Lane, 1983). Assume the reduced model (10.26) under the null; with its estimates $\hat{\boldsymbol{\beta}}_{0(0)} = (\hat{\beta}_{0(0)}, \ldots, \hat{\beta}_{k(0)})$ and residuals $(e_{1(0)}, \ldots, e_{n(0)})' \equiv \mathbf{e}_0 = (I_n - H_0)\mathbf{y}$, build the most plausible null model

$$y_i = \hat{\beta}_{0(0)} + \hat{\beta}_{1(0)} x_{i1} + \cdots + \hat{\beta}_{k(0)} x_{ik} + e_{i(0)}, \qquad i = 1, \ldots, n. \tag{10.27}$$

In this approximate model, shuffle the (approximately exchangeable) residuals $\{e_{1(0)}, \ldots, e_{n(0)}\}$ by sampling without replacement to $(e_{1(0)}^*, \ldots, e_{n(0)}^*)$ and form the shuffled responses

$$y_i^* = \hat{\beta}_{0(0)} + \hat{\beta}_{1(0)} x_{i1} + \cdots + \hat{\beta}_{k(0)} x_{ik} + e_{i(0)}^*, \qquad i = 1, \ldots, n, \tag{10.28}$$

and finally $\mathcal{D}^* = \{(\mathbf{x}_1, y_1^*), \ldots, (\mathbf{x}_n, y_n^*)\}$. The procedure is the same as steps 1–3 above; only the generation of $\mathcal{D}^*$ differs.

Shuffling the residuals of the full model. The third method is the permutation version of bootstrap resampling with a pivot. The two methods above produce shuffled samples plausible under the null; this one produces samples plausible under the alternative (when it is true). It shuffles the full-model residuals, whose variance is smaller than that of the reduced-model residuals, aiming at a less dispersed null distribution of F (i.e. higher power) (Ter Braak, 1992). Get the full-model residuals $(e_1, \ldots, e_n)' \equiv \mathbf{e} = (I_n - H)\mathbf{y}$ (adjusted residuals r_i (10.18) may be used), shuffle them without replacement to $(e_1^*, \ldots, e_n^*)$, add them to the full-model fitted values $\hat{y}_i$:

$$y_i^* = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip} + e_i^* = \hat{y}_i + e_i^*, \qquad i = 1, \ldots, n, \tag{10.29}$$

and form $\mathcal{D}_{\text{full}}^* = \{(\mathbf{x}_1, y_1^*), \ldots, (\mathbf{x}_n, y_n^*)\}$. This is a sample from the full (alternative) model: the least squares estimate $\hat{\boldsymbol{\beta}}^*$ from $\mathcal{D}_{\text{full}}^*$ lies "near" $\hat{\boldsymbol{\beta}}$, like a bootstrap replicate. (From samples shuffled under the null, $\hat{\beta}_{k+1}^*, \ldots, \hat{\beta}_p^*$ are near 0.) So with this shuffling the statistic must be expressible as a pivot.

The usual F follows the F-distribution only when $\boldsymbol{\beta}_1 = 0$ (under normality), so expressing it as a pivot for $\boldsymbol{\beta}_1 \neq 0$ seems hard. But knowing that $SSR_{\beta_1|\beta_0}$ in (10.25) equals $\hat{\boldsymbol{\beta}}_1'\{X_1'(I_n - H_0)X_1\}\hat{\boldsymbol{\beta}}_1$,[4]

$$F = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)} = \frac{(\hat{\boldsymbol{\beta}}_1 - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - 0)}{q \cdot s^2}$$

[4] Accept this at the level of the book. (Supplied: by the Frisch–Waugh–Lovell theorem, $\hat{\boldsymbol{\beta}}_1$ is the least squares coefficient of $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ is the projection onto $\text{col}(\tilde{X}_1)$, so $\mathbf{y}'(H - H_0)\mathbf{y} = \hat{\boldsymbol{\beta}}_1'\tilde{X}_1'\tilde{X}_1\hat{\boldsymbol{\beta}}_1$.)

(s^2 the full-model error variance estimate (10.16)). So

$$F_{\beta_1} = \frac{(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1)}{q \cdot s^2} \tag{10.30}$$

follows the same $F_{q,n-p-1}$ distribution even when $\boldsymbol{\beta}_1 \neq 0$: a pivot. Even when F_{β_1} is not $F_{q,n-p-1}$ (normality cannot be assumed), if it is a pivot, the permutation test with the shuffled samples $\mathcal{D}^*_{\text{full}}$:

1. Compute F_{obs} (10.25) from $\mathcal{D}$.
2. For $i = 1, \ldots, M$ ($M = 1999$) generate a randomly shuffled sample $\mathcal{D}^{*i}_{\text{full}}$, compute $\hat{\boldsymbol{\beta}}_1^{*i}$ and $(s^{*i})^2$, and
$$F^{*i}_{\beta_1} = \frac{(\hat{\boldsymbol{\beta}}_1^{*i} - \hat{\boldsymbol{\beta}}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*i} - \hat{\boldsymbol{\beta}}_1)}{q \cdot (s^{*i})^2}.$$
3. The p-value for (10.22) is $p \approx \sum_{i=0}^{M} I(F^{*i}_{\beta_1} \geq F_{\text{obs}})/(M+1)$ ($F^{*0}_{\beta_1} := F_{\text{obs}}$).

Remark

A hypothesis on one coefficient can also be tested with F. But for the one-sided test

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j > 0 \tag{10.31}$$

the t-statistic $t_j = (\hat{\beta}_j - \beta_j)/\widehat{\text{se}}(\hat{\beta}_j)$ is better, and all three permutation methods above can be used. E.g. with the pivot t_j and full-model residual shuffling:

1. Compute $t_{j(0)} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j)$ from $\mathcal{D}$.
2. For $i = 1, \ldots, M$ ($M = 1999$) generate $\mathcal{D}^{*i}_{\text{full}}$ (full-model residual shuffling), compute $\hat{\beta}_j^*$, $\widehat{\text{se}}(\hat{\beta}_j^*)$ and
$$t_j^{*i} = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^*), \qquad i = 1, \ldots, M. \tag{10.32}$$
3. The p-value for (10.31) is $p \approx \sum_{i=0}^{M} I(t_j^{*i} \geq t_{j(0)})/(M+1)$ ($t_j^{*0} := t_{j(0)}$).

Example 10.10

For $H_0 : \beta_5 = \beta_6 = 0$ of Example 10.9, the three random permutation tests. First a function `F.pivot()` computes F in the pivot form (10.30):

```
> X10 = X1 - Hat0 %*% X1
> XX10 = t(X10) %*% X10
> XXX10 = solve( t(X10) %*% X10) %*% t(X10)
> F.pivot = function(y.perm, beta1){
+   beta1.hat = XXX10 %*% y.perm
+   SSE = sum(y.perm^2) - t(y.perm) %*% Hat %*% y.perm
+   SSR = t(beta1.hat - beta1) %*% XX10 %*% (beta1.hat - beta1)
+   as.vector(SSR / SSE / q * (n-p-1))
+ }
```

The design matrix is fixed under every shuffle, so for speed H, H_0, X_0, $X_{10} := (I_n - H_0)X_1$, $X_{10}'X_{10} = X_1'(I_n - H_0)X_1$ and $(X_{10}'X_{10})^{-1}X_{10}'$ are computed once outside the function. The observed F_{obs} uses the original response vector and the null value $(\beta_5, \beta_6) = (0, 0)$; as in Example 10.9 it is 3.599.

```
> (F.obs = F.pivot(y, c(0,0)))
[1] 3.599035
```

(1) Shuffling the observations: sample the response vector without replacement.

```
F.perm.raw = replicate(M, {
  yperm = sample(y,n, replace = FALSE)
  F.pivot(yperm, c(0,0))
})
p.raw = mean(c(F.obs,F.perm.raw) >= F.obs)
```

(2) Shuffling the null residuals needs the reduced-model fitted values $\hat{\mathbf{y}}_0$ and residuals $\mathbf{e}_0$.

```
yhat0 = Hat0 %*% y
res0 = y - yhat0
F.perm.null = replicate(M, {
  yperm = yhat0 + sample(res0,n, replace = FALSE)
  F.pivot(yperm, c(0,0))
})
p.null = mean(c(F.obs,F.perm.null) >= F.obs)
```

(3) Shuffling the full-model residuals needs the fitted values $\hat{\mathbf{y}}$, residuals $\mathbf{e}$, and, for the pivot, the full-model $\hat{\boldsymbol{\beta}}_1$.

```
yhat = Hat %*% y
res = y - yhat
beta.hat1 = XXX10 %*% y
F.perm.full = replicate(M, {
  yperm = yhat + sample(res,n, replace = FALSE)
  F.pivot(yperm, beta.hat1)
})
p.full = mean(c(F.obs,F.perm.full) >= F.obs)
```

The p-values are all similar:

```
> c(p.theory, p.raw, p.null, p.full)
[1] 0.02980354 0.03150000 0.02850000 0.03300000
```

For these data the three permutation distributions are also nearly the same (histograms in the book, compared with the theoretical $F_{q,n-p-1}$ and F_{obs}).

Comparing the three methods. All three use the explanatory values X (in particular X_0 of the reduced model) as they are, so all are tests conditional on the explanatory values. Shuffling the observations is known not to control the type I error in small samples (type I error above α). It actually generates the permutation distribution under $H_0 : \beta_1 = \cdots = \beta_p = 0$, not $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$, so it is not appropriate. The null-residual and full-residual shuffles draw residuals by simple sampling without replacement; drawing with replacement instead gives bootstrap resampling. (Two of the bootstrap tests in Section 10.3.3 are these.) Finally, the full-model resampling hinges on the existence of a pivot.

The three methods are asymptotically equivalent for large n; for finite n simulations show Freedman and Lane's (1983) shuffling of the null (reduced-model) residuals is best.[5]

10.3.2 Bootstrap confidence intervals for the coefficients

Bootstrap intervals for β_j in multiple regression proceed almost as in simple regression. Bivariate bootstrap could be done by the case or residual bootstrap. Multiple regression data have $p + 1$ variables, but the p explanatory variables play the role of "X" in bivariate data, so the extension is natural:

[5] See Anderson and Robinson (2001); Anderson and Legendre (1999).

- **Case bootstrap:** resample the n pairs $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ with replacement.
- **Residual bootstrap:** from the residuals $e_i = y_i - \hat{y}_i$ or adjusted residuals $\{r_1, \ldots, r_n\}$ (10.18), draw $r_1^*, \ldots, r_n^*$ with replacement and form $(\mathbf{x}_i^*, y_i^*) = (\mathbf{x}_i, \hat{y}_i + r_i^*)$.

Pros and cons are as in simple regression, but with the residual bootstrap the design matrix X (and H) is the same in every bootstrap sample, so the computational advantage is larger. With bootstrap replicates of $\hat{\beta}_j$ the bias, standard error and intervals of β_j can be estimated. When n is much larger than p, bootstrap samples approximate the population well and the estimates are reliable.

The code below uses the residual bootstrap to compute X, $(X'X)^{-1}$ (`XX`) and H (`Hat`) once and gives 95% bootstrap-t intervals for all β_j; the pivot is $t_j = (\hat{\beta}_j - \beta_j)/\widehat{\text{se}}(\hat{\beta}_j)$ (10.20).

Example 10.11 (Residual bootstrap, bootstrap-t intervals)

Continuing Example 10.10, to get bootstrap-t intervals for the coefficients, compute $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^*)$ for all $j = 0, 1, \ldots, p$ in each bootstrap sample; define `t.pivot()` first:

```
XX = solve(t(X) %*% X)
Hat = X %*% XX %*% t(X)
t.pivot = function(yboot, beta){
          yhat = Hat %*% yboot
          res = yboot - yhat
          beta.hat  = XX %*% t(X) %*% yboot
          se.beta.hat = sqrt( sum(res^2) / (n-p-1) * diag(XX))
          as.vector( (beta.hat - beta)  / se.beta.hat )
}
```

The bootstrap replicates t_j^*: except for using adjusted residuals, this is almost the same as the shuffling of full-model residuals in the permutation test; the only difference is the `replace` argument of `sample()` (`TRUE` for the bootstrap, `FALSE` for permutation).

```
yhat = lm.out$fitted.values
adjusted.res = lm.out$residuals / sqrt(1 - hatvalues(lm.out))
beta.hat = lm.out$coefficients
B = 1999
t.boot.res = replicate(B, {
        yboot = yhat + sample(adjusted.res, n, replace = TRUE)
        t.pivot(yboot, beta.hat)
}) %>% t()
```

The 95% bootstrap-t interval for β_5 (Generosity) is below; these data roughly satisfy normality, so it is almost the same as the normal interval of Example 10.8.

```
> ci.beta5 = beta.hat[6] - se.hat[6] * quantile(t.boot.res[,6],
    probs = c(1-0.025,0.025))
> ci.beta5
     97.5%      2.5%
-0.3737554  1.6320302
```

10.3.3 Bootstrap tests

In simple regression there were four bootstrap tests for β: two resamplings (case and residual bootstrap) under two scenarios (resampling under the null, resampling with a pivot). Tests in multiple regression proceed the same way. For $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$ use the pivot F (10.30);

for $H_0 : \beta_j = 0$ the pivot t (10.20). The observed statistics are

$$F_{\text{obs}} = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)} = \frac{(\hat{\boldsymbol{\beta}}_1 - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - 0)}{q \cdot s^2} \tag{10.33}$$

or

$$t_{j(\text{obs})} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j). \tag{10.34}$$

To approximate their null distributions by resampling under the null:

- **Resampling under the null, case bootstrap.** Draw the explanatory bootstrap sample $\mathbf{x}_1^*, \ldots, \mathbf{x}_n^*$ from $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ with replacement and, independently, the responses $y_1^*, \ldots, y_n^*$ from $\{y_1, \ldots, y_n\}$; form $\{(\mathbf{x}_1^*, y_1^*), \ldots, (\mathbf{x}_n^*, y_n^*)\}$.
- **Resampling under the null, residual bootstrap.** With the null (reduced) model fitted values $\hat{y}_0$ and residuals $r_{i(0)}$, draw the residuals with replacement and form $y_i^* = \hat{y}_{i(0)} + r_{i(0)}^*$, giving $\{(\mathbf{x}_1, y_1^*), \ldots, (\mathbf{x}_n, y_n^*)\}$. (The resampled observations are almost those of shuffling the null residuals (10.28); only sampling is with replacement.)

In each bootstrap sample ($b = 1, \ldots, B$) compute

$$F^{*b} = \frac{(\hat{\boldsymbol{\beta}}_1^{*b} - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*b} - 0)}{q \cdot (s^{*b})^2}$$

or $t_j^{*b} = (\hat{\beta}_j^{*b} - 0)/\widehat{\text{se}}(\hat{\beta}_j^{*b})$, and the two-sided p-values ($F^{*0} := F_{\text{obs}}$, $t_j^{*0} = t_{j(\text{obs})}$)

$$p_F^* = \sum_{b=0}^{B} I(F^{*b} \geq F_{\text{obs}})/(B+1), \qquad p_t^* = \sum_{b=0}^{B} I(|t_j^{*b}| \geq |t_{j(\text{obs})}|)/(B+1). \tag{10.35}$$

To approximate the null distribution with a pivot:

- **Pivot, case bootstrap.** Resample the n pairs $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ with replacement (as for intervals).
- **Pivot, residual bootstrap.** From the residuals $e_i = y - \hat{y}_i$ or adjusted residuals $\{r_1, \ldots, r_n\}$ (10.18) draw $r_1^*, \ldots, r_n^*$ with replacement and form $(\mathbf{x}_i^*, y_i^*) = (\mathbf{x}_i, \hat{y}_i + r_i^*)$ (as for intervals; also as in shuffling the full-model residuals (10.29) but with replacement).

In each bootstrap sample compute

$$F^{*b} = \frac{(\hat{\boldsymbol{\beta}}_1^{*b} - \hat{\boldsymbol{\beta}}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*b} - \hat{\boldsymbol{\beta}}_1)}{q \cdot (s^{*b})^2}$$

or $t_j^* = (\hat{\beta}_j^{*b} - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^{*b})$, and the two-sided p-value by (10.35). The pivot-based bootstrap test of $H_0 : \beta_j = 0$ is the inversion of the bootstrap-t interval for β_j.

Example 10.12

In Example 10.11 the 95% bootstrap-t interval for β_5 (Generosity) contained 0. Test $H_0 : \beta_5 = 0$ against $H_1 : \beta_5 \neq 0$ by the pivot-based residual bootstrap; the replicates (10.32) were already stored in `t.boot.res` in Example 10.11, so:

```
> t5 = t.pivot(y, rep(0,p+1))[6]
> (pvalue = mean(abs( c(t5, t.boot.res[,6])) >= abs(t5)))
[1] 0.215
```

Compared with the 95% bootstrap-t interval $(-0.39, 1.614)$ from `ci.beta5`: the null value 0

is in the interval, so H_0 is not rejected at the 5% level.

10.4 Robust linear regression

So far, permutation tests and bootstrap resampling in simple and multiple regression assumed least squares estimation of the coefficients. The least squares estimator coincides with the maximum likelihood estimator when the errors are independent and identically normal, so in the two situations

- the error distribution has much heavier or skewed tails than normal,
- the observations contain outliers,

least squares loses efficiency.[6] Then nonparametric or robust estimators are preferred.

This section covers nonparametric, robust estimators of regression coefficients. "Nonparametric" means estimators using ranks and signs in the traditional sense; "robust" means estimators like the median and trimmed means whose value does not change much when part of the data is contaminated. Estimators using ranks and signs keep the same ranks and signs even if the largest value (e.g. 10) is contaminated to 10^{100}, so they are a kind of robust estimator. We first introduce robust estimators of the location of one population to understand the concept, then apply them to regression.

10.4.1 Robust estimation of the mean

Introducing permutation tests, we used not only the difference of sample means but also of medians or trimmed means as statistics for the location difference, and saw that the efficiency (power) depends on the statistic. For asymmetric or heavy-tailed populations, medians and trimmed means were more powerful than means. Heavy-tailed distributions often produce values far from the center, which are often regarded as outliers. (Conversely, data with outliers can be viewed as sampled from a very heavy-tailed distribution.) For the location (the population mean, when it is defined), the sample mean is very sensitive to outliers, a "non-robust" statistic; the median and trimmed mean are representative robust statistics. There are many measures of robustness; here we use a high *breakdown point* as the criterion.

Breakdown point. For a statistic $T = T(\mathbb{X}_n)$ computed from $\mathbb{X}_n = \{x_1, \ldots, x_n\}$: if sending one observation x_i to infinity sends T to infinity ($\lim_{x_i \to \infty} T(\mathbb{X}_n) = \infty$), the statistic "breaks down." If sending any $b-1$ observations to infinity does not break it, but sending some b observations $x_{i_1}, \ldots, x_{i_b}$ to infinity makes it infinite, it breaks when b or more observations are contaminated. The breakdown point of T is the smallest fraction of contaminated observations that breaks it: increasing the number of contaminations from 1, if for any $b-1$ observations

$$\lim_{x_{i_1}, \ldots, x_{i_{b-1}} \to \infty} |T(\mathbb{X}_n)| < \infty$$

while for some b observations

$$\lim_{x_{i_1}, \ldots, x_{i_b} \to \infty} |T(\mathbb{X}_n)| = \infty,$$

[6] Loss of efficiency of $\hat{\beta}$ means all of: (1) $\hat{\beta}$ has large MSE, variance or standard error; (2) intervals for β based on $\hat{\beta}$ are long; (3) tests have low power. If normality fails, normal-theory tests do not control the type I error, but permutation (or bootstrap) tests based on $\hat{\beta}$ do (approximately in multiple regression); their power can still be low. (In two-sample comparisons, the permutation test with the two-sample t was less powerful than with the difference of medians.) Likewise, bootstrap intervals based on $\hat{\beta}$ are long but (approximately) correct.

the breakdown point is

$$BP(T) = \frac{b}{n}.$$

For the sample mean, median and 10% trimmed mean, roughly

$$BP(\bar{X}_n) = \frac{1}{n} \approx 0, \qquad BP(\text{median}\{X_i\}) = \frac{1}{2}, \qquad BP(10\% \text{ trimmed mean}) = 10\%,$$

easily seen. The median, with breakdown point 50%, is the most robust estimator. (Nothing is free: most robust estimators lose efficiency. Example 2.16 showed the ARE of the sample median to the sample mean is 63.7%.)

The simplest way to devise robust estimators is to "ignore" observations far from the "center." The median is computed from the middle one or two observations, ignoring all others. The 10% trimmed mean removes the 10% of observations at each extreme and averages the remaining 80%. Replacing them instead by the 10% and 90% sample quantiles and averaging gives the Winsorized mean; the 10% Winsorized mean has breakdown point 10%.

Example 10.13

Suppose $n = 11$ observations $-5, -3, -3, -3, -1, 0, 1, 2, 3, 4, 1000$ are sampled from a population; the last, $x_{11} = 1000$, is a contaminated outlier. The mean, 10% trimmed mean and 10% Winsorized mean (R package `WRS2`):

```
> x = c(-5,-3,-3,-3,-1,0,1,2,3,4,1000)
> c(mean(x), mean(x, tr = 0.1), WRS2::winmean(x, tr = 0.1))
[1] 90.45454545  0.00000000  0.09090909
```

The trimmed mean averages the 9 middle observations after removing -5 and 1000. The Winsorized mean replaces -5 by the 10% sample quantile -3 and 1000 by the 90% quantile 4, then averages all 11 (Winsorized) observations. Substituting 10^{100} for the outlier 1000 does not change the trimmed or Winsorized means.

M-estimators. Trimmed and Winsorized means remove the same fraction of observations at both extremes. If, as in Example 10.13, the outliers are only on the right, there is no reason to remove or replace the normal observation -5. The M-estimators introduced here take the distribution of the observations into account and remove or adjust only the effect of the outliers, wherever they are.

First another expression for $\bar{x}_n$. If $x_i = \mu + \epsilon_i$, $i = 1, \ldots, n$, with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon$ of mean 0, the estimator $\bar{x}_n$ of μ is the least squares estimator:

$$\bar{x}_n = \arg\min_\mu \sum_{i=1}^n (x_i - \mu)^2 = \arg\min_\mu \sum_{i=1}^n \epsilon_i^2. \tag{10.36}$$

($\arg\min_\mu$ is the μ minimizing $\sum (x_i - \mu)^2$.) Differentiating in μ, the minimizer solves $\sum_{i=1}^n (x_i - \mu) = 0$, i.e. $\hat{\mu} = \bar{x}_n$. The trimmed and Winsorized means are least squares estimators computed after removing some observations with large $|x_i - \mu|$ (trimmed) or replacing some x_i by other values (Winsorized), on the grounds that observations with large $|x_i - \mu|$ are likely outliers. The sample mean is not robust because for a large outlier x_i its squared error $(x_i - \mu)^2$ is huge, and reducing it pulls $\bar{x}_n$ strongly toward the outlier. The key idea of M-estimators of μ is to use, instead of $(x_i - \mu)^2$, a function that adjusts the weight of the error so that an observation with large $|x_i - \mu|$ can be suitably "ignored."

Remark

If the errors are normal, $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, a parametric model, and the log-likelihood is

$$\ell(\mu; \mathbb{X}_n) = -\frac{1}{2}\sum_{i=1}^{n}\Big(\frac{x_i - \mu}{\sigma}\Big)^2 + \text{const.} \tag{10.37}$$

With σ^2 known, the maximum likelihood estimator of μ,

$$\bar{x}_n = \arg\max_{\mu} \ell(\mu; \mathbb{X}_n) = \arg\min_{\mu} \sum_{i=1}^{n}\Big(\frac{x_i - \mu}{\sigma}\Big)^2, \tag{10.38}$$

is the least squares estimator. The "M" of M-estimator means an estimator of "MLE type." It may also be read as "an estimator of minimum (or maximum) type."

An M-estimator replaces the square function $y = x^2$ by some function $y = \Psi(x)$:

$$\hat{\mu}^{\Psi} = \arg\min_{\mu} \sum_{i=1}^{n} \Psi\Big(\frac{x_i - \mu}{\tau}\Big),$$

where τ is an estimator of the population standard deviation σ, possibly estimated from the data. Equivalently, the M-estimator $\hat{\mu}^{\Psi}$ is the solution of

$$\sum_{i=1}^{n} \psi\Big(\frac{x_i - \mu}{\tau}\Big) = 0$$

with $\psi(x) = \frac{d}{dx}\Psi(x)$ replacing the derivative $\frac{d}{dx}x^2 = 2x$ of the square function. The M-estimator is determined by Ψ (or ψ). Representative examples:

1. The sample mean: the M-estimator with $\Psi(x) = x^2/2$, i.e. $\psi(x) = x$.
2. The sample median: the M-estimator with $\Psi(x) = |x|$, $\psi(x) = \text{sgn}(x)$.
3. Huber's (1964) Ψ: for some $k > 0$, $\psi(x) = \max\{-k, \min\{k, x\}\}$, and

$$\Psi(x) = \begin{cases} x^2/2, & |x| \le k; \\ k|x| - \frac{k^2}{2}, & |x| > k. \end{cases}$$

   The constant k can be set to various values, but generally $k = 1.28$, the 90% quantile of the standard normal, is used.

4. Tukey's "biweight" function: for some $c > 0$,

$$\psi(x) = \begin{cases} x(1 - (\frac{x}{c})^2)^2, & |x| \le c; \\ 0, & |x| > c. \end{cases}$$

(Figure 10.4 of the book plots these ψ.) Except for the sample mean, $|\psi(x)|$ is always below a fixed constant. Since the input x of ψ is the standardized error $(x_i - \mu)/\tau$, a large error does not make $|\psi(x)|$ large, so very large errors are suitably ignored. Huber's Ψ has the form of the mean's Ψ in the "center" ($|x| \le k$) and of the median's in the "periphery"; this form was designed to have both the high efficiency of the mean and the high robustness of the median. Huber's M-estimator is the solution in μ of

$$\sum_{i=1}^{n} \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi(x) = \max\{-k, \min\{k, x\}\}, \tag{10.39}$$

where $\hat{\tau} = \text{MAD}\{x_1, \ldots, x_n\} \times 1.4826$ is a robust estimator of σ under the normal $N(\mu, \sigma^2)$.[7] (Supplied: for $N(\mu, \sigma^2)$, MAD $\to \sigma\Phi^{-1}(3/4) \approx 0.6745\sigma$, and $1/0.6745 \approx 1.4826$.) Equation (10.39) has no analytic solution, so Huber's M-estimator $\hat{\mu}^{\Psi}$ is found numerically. The M-estimator with Huber's Ψ is the most representative robust location estimator.

Example 10.14

The data of Example 10.13 seem to have the outlier 1000. For Huber's M-estimator of location use `mest()` of the `WRS2` package ("M-estimation"); `bend = 1.28` sets $k = 1.28$.

```
> WRS2::mest(x,bend = 1.28)
[1] 0.0693184
```

The standardized form $(1000 - \hat{\mu}^{\Psi})/\hat{\tau}$ of the last observation was Winsorized; the other observations were used as they are in computing the mean.

The breakdown point of an M-estimator is always 50% when $|\psi(x)|$ is bounded. The MAD used in Huber's M-estimator also has breakdown point 50%, so $BP(\hat{\mu}^{\Psi}) = 50\%$. The efficiency of Huber's M-estimator depends on the choice of k: in normal populations it increases with k, and at $k = 1.35$ the ARE of Huber's M-estimator to the sample mean is known to be about 90%.

10.4.2 Inference on location with outliers

When outliers are suspected, how should intervals for a location parameter be estimated and hypotheses tested? Assume observations $X_i = \mu + \epsilon_i$, $E(\epsilon_i) = 0$, but with contaminated data.

Example 10.15

The `phones` data of R's `MASS` package record the number of telephone calls (millions) over the 24 years 1950–1973 (Figure 10.5 of the book). At least 6 of the 24 observations look like outliers. Using these data we want an interval for the mean number of calls in the 50s–70s and a test of whether the mean is 15 (million).

The hypotheses of Example 10.15 are $H_0 : \mu = 15$ and $H_1 : \mu \neq 15$. From Figure 10.5, removing the 6 apparent outliers one would expect not to reject $\mu = 15$. If the uncontaminated observations X_i are assumed symmetric, the sign test and signed-rank test of Chapter 2 are good methods. The next example confirms that traditional distribution-free methods work for contaminated data.

Example 10.16

With the data of Example 10.15, the t-statistic $\sqrt{n}(\bar{X}_n - \mu_0)/s$ is pulled up by the outliers (the sample mean $\bar{X}_n$ is large), and $H_0 : \mu = 15$ is rejected at 5%. The t-statistic is not free of the influence of outliers, so it is a poor test here.

```
> x = MASS::phones$calls
> mean(x)
[1] 49.99167
> t.test(x, mu = 15)$p.value
[1] 0.0154505
```

The large-sample p-value of the signed-rank test (Section 2.2) is 0.1573, not significant at 5%, and the large-sample interval for μ is $(12.7, 81.6)$, containing 15. With the signed-rank test $H_0 : \mu = 15$ is not rejected.

```
> wilcox.test(x,mu = 15, alternative = "two.sided",
```

[7] R's `mad()` computes not $\text{MAD}\{x_1, \ldots, x_n\}$ but $\text{MAD}\{x_1, \ldots, x_n\} \times 1.4826$.

```
+             exact = F, conf.int = T)

        Wilcoxon signed rank test with continuity correction

data:  x
V = 200, p-value = 0.1573
alternative hypothesis: true location is not equal to 15
95 percent confidence interval:
 12.74998 81.59997
sample estimates:
(pseudo)median
        24.306
```

The exact p-value of the sign test (Section 2.1) is 0.84, and the sign-test interval is $(8.1, 43)$: the same conclusion as the signed-rank test.

```
> s = sign(x-15)
> n = sum(s != 0)
> (Sobs = sum(s > 0))
[1] 12
> (pvalue.sign.test = 2*(1- pbinom(Sobs,n,1/2)))
[1] 0.8388197
>
> alpha = 0.05
> l = qbinom(alpha/2, size = n, prob = 1/2); u = n+ 1 - l
> (ci.sign.test.x = sort(x)[c(l,u)]) # CI by sign test
[1]  8.1 43.0
```

(The book's two-sided sign p-value code computes $2(1-P(S \leq 12))$; the exact two-sided value is $2\min\{P(S \geq 12), P(S \leq 12)\}$, which leads to the same conclusion.) The Hodges–Lehmann estimate of the signed-rank test, the median of the Walsh averages, 24.306, is closer to the sign test's Hodges–Lehmann estimate, the sample median 15.5, than to the (contaminated) sample mean 49.99. The breakdown point of the sample mean is 0, of the median of the Walsh averages about 29%, and of the median 50%; the lower the breakdown point, the less robust, and the interval from a low-breakdown statistic is wider.

Breakdown point of the Hodges–Lehmann estimator (supplied)

If m observations are sent to infinity, the Walsh averages involving them also go to infinity; their number is about $n^2/2-(n-m)^2/2$ of $\approx n^2/2$. The median of the Walsh averages breaks down when this fraction exceeds $1/2$, i.e. $1-(1-m/n)^2 > 1/2$, i.e. $m/n > 1-1/\sqrt{2} \approx 0.293$.

Next, intervals from Huber's M-estimator. M-estimators including Huber's are asymptotically normal for large samples, so if the standard error can be estimated, normal-approximation intervals and tests are possible.[8] Below, from Huber's M-estimator, the 95% interval for μ by (1) the normal-approximation bootstrap interval with bootstrap estimates of standard error and bias (Section 8.1), and (2) the percentile bootstrap interval (Section 8.4).

Example 10.17

From the data of Example 10.15, Huber's M-estimator is $\hat{\mu}^{\Psi} = 20.84$.

```
> (mest.obs = WRS2::mest(x))
[1] 20.84105
```

[8] Estimating the standard error of M-estimators requires the influence function, a concept not introduced in this book, so it is not treated in detail. Instead the bootstrap can be used to estimate the standard error.

Bootstrap approximation of its bias and standard error:

```
> set.seed(1)
> B = 1999
> mest.boot = replicate(B,WRS2::mest(sample(x, length(x),
    replace = T)))
> (se.est = sd(mest.boot))
[1] 11.46062
> (bias.est = mean(mest.boot) - mest.obs)
[1] 2.514652
```

For large samples the M-estimator is normal, so

$$\frac{\hat{\mu}^{\Psi} - \text{bias}(\hat{\mu}^{\Psi}) - \mu}{\text{se}(\hat{\mu}^{\Psi})} \dot{\sim} N(0,1);$$

plugging in the bootstrap estimates of bias and standard error gives the 95% normal-approximation bootstrap interval:

```
> (norm.app.boot.CI = mest.obs - bias.est + se.est * qnorm
    (0.025) *c(1,-1))
[1] -4.136005 40.788792
```

The interval for μ, which must be positive, contains negative values: the sample is small and the normal approximation is poor. The percentile bootstrap interval seems more accurate:

```
> (boot.perc.ci.mest = quantile(mest.boot, probs = c
    (0.025,1-0.025)))
    2.5%    97.5%
11.24107 53.34024
```

For reference, the 95% BCa interval for μ from the same bootstrap resampling is $(12.1, 67.1)$.

10.4.3 Robust estimators of regression coefficients

On the same principle as robust estimators of location for contaminated data, various robust estimators have been developed for regression coefficients.

Rank tests and the Theil–Sen estimator. The most traditional robust estimator of regression coefficients using ranks and signs is the Theil–Sen estimator (Theil, 1950; Sen, 1968), defined for the coefficients α, β of the simple regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon. \tag{10.40}$$

Theil defined it with ranks; Sen showed it is related to Kendall's tau. From each pair of observations (x_i, y_i), (x_j, y_j) of $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ compute the "slope"

$$\beta_{ij} := \frac{y_i - y_j}{x_i - x_j}, \qquad 1 \le i < j \le n; \tag{10.41}$$

their median

$$\hat{\beta}^{TS} = \text{median}\{\beta_{ij} : 1 \le i < j \le n\} \tag{10.42}$$

is the Theil–Sen estimator of β (pairs with $x_i = x_j$ excluded). This resembles the median of the Walsh averages (from the signed-rank test for one location) or of the Mann–Whitney–Wilcoxon differences used as estimators. The Theil–Sen estimator of α is

$$\hat{\alpha}^{TS} = \text{median}\{y_i - \hat{\beta}^{TS} x_i\}, \tag{10.43}$$

or the median of the Walsh averages of the $y_i - \hat{\beta}^{TS} x_i$ (here simply (10.43)).

To see the relation to Kendall's tau, recall the test with Kendall's tau statistic. If model (10.40) is true and we test $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$, under the null the two variables are independent, so the test with Kendall's τ_n (exact permutation test or large-sample approximation) is possible. With $K_n(\cdot)$ the distribution function of the null distribution of Kendall's tau, $p = 2\min\{1 - K_n(\tau_n), K_n(\tau_n)\}$. How do we test, for a given slope β_0,

$$H_0 : \beta = \beta_0 \quad \text{vs} \quad H_1 : \beta \neq \beta_0? \tag{10.44}$$

Under the null $y_i = \alpha + \beta_0 x_i + \epsilon_i$, so the pairs $(\alpha + \epsilon_i, x_i) = (y_i - \beta_0 x_i, x_i)$ are independent, and Kendall's statistic is $\tau_n(\beta_0) = \sum_{i<j} U_{ij}(\beta_0) / \binom{n}{2}$ with

$$\begin{aligned} U_{ij}(\beta_0) &= \operatorname{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] \\ &= \operatorname{sgn}[(x_i - x_j)\{(y_i - y_j) - \beta_0 (x_i - x_j)\}] \\ &= \operatorname{sgn}[(x_i - x_j)^2 (\beta_{ij} - \beta_0)] \\ &= \operatorname{sgn}(\beta_{ij} - \beta_0), \end{aligned}$$

with β_{ij} as in (10.41). Two facts: first, if $H_0 : \beta = \beta_0$ is true, the null distribution of $\tau_n(\beta_0)$ is again K_n, so the test at a given level is possible. Second, Kendall's statistic is

$$\tau_n(\beta_0) = \frac{1}{\binom{n}{2}} \big[\#\{\beta_{ij} - \beta_0 > 0\} - \#\{\beta_{ij} - \beta_0 < 0\} \big], \tag{10.45}$$

so the result depends on the position of β_0 relative to the pairwise slopes $\{\beta_{ij}\}$. By the same logic as the interval from the order statistics of the Walsh averages for the signed-rank test, the interval for β from Kendall's tau uses the order statistics $\beta_{(1)} < \cdots < \beta_{(N)}$ $(N = \binom{n}{2})$ of the slopes. Under the null the distribution of $\tau_n(\beta_0)$ is always Kendall's tau distribution, so with $Q_K(p; n)$ its p-quantile, the $1 - \alpha$ interval for β is $(\beta_{(l)}, \beta_{(u)})$ with

$$l = \binom{n}{2} \cdot \frac{Q_K(\alpha/2, n) + 1}{2}, \qquad u = \binom{n}{2} \cdot \frac{Q_K(1 - \alpha/2, n) + 1}{2}. \tag{10.46}$$

Justification of (10.46) (supplied; Exercise 10.6). With no ties, $N\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\} = 2\#\{\beta_{ij} > \beta_0\} - N$. The level-$\alpha$ test accepts $H_0 : \beta = \beta_0$ iff $Q_K(\frac{\alpha}{2}) \leq \tau_n(\beta_0) \leq Q_K(1 - \frac{\alpha}{2})$ (with the usual care at the discrete boundary), i.e. iff

$$N \frac{1 + Q_K(\frac{\alpha}{2})}{2} \leq \#\{\beta_{ij} > \beta_0\} \leq N \frac{1 + Q_K(1 - \frac{\alpha}{2})}{2}.$$

$\#\{\beta_{ij} > \beta_0\}$ is a decreasing step function of β_0 that drops by one at each $\beta_{(m)}$, so the acceptance set is an interval of consecutive order statistics of the slopes. Its endpoints are the order statistics with indices given by (10.46) (up to the symmetric relabelling $m \leftrightarrow N + 1 - m$, using the symmetry of K_n). Since $\tau_n(\beta) \sim K_n$ at the true β, the coverage is at least $1 - \alpha$. ■

As the level of this interval is lowered, what remains is the median of the pairwise slopes, the Theil–Sen slope estimator $\hat{\beta}^{TS}$ (10.42). The Theil–Sen estimator is intuitive and easy to explain, has a high breakdown point $BP(\hat{\beta}^{TS}) = 29.3\%$, and is known to have about 96% ARE to least squares under normal errors (Wilcox, 2011; Peng et al., 2008).

Example 10.18 (Theil–Sen estimator in simple regression)

For the GDP and happiness data of 13 countries in Example 5.2, the least squares estimates are $\hat{\alpha} = 6.5$, $\hat{\beta} = 0.6$: the relation between GDP and happiness does not look strong. Under normality the usual t-test of $H_0 : \beta = 0$ has $p = 0.311$, so the relation is not significant.

But one data point has an unusually large GDP that seems to distort the overall trend, and the independence test with Kendall's tau (large-sample approximation) gives $p = 0.02$: significant. The Theil–Sen test of the same null obviously gives the same $p = 0.02$. The R code below tests independence by Kendall's tau:

```
> wh = read.csv("data/whappiness.csv")
> x = wh$GDP.per.capita
> y = wh$Score
> cor.test(x,y, method = "k")$p.value
[1] 0.01713551
```

Now compare the least squares line with the Theil–Sen line. The Theil–Sen estimates:

```
> n = length(x)
> pairids = gtools::combinations(n,2)
> pairbetas = data.frame(i = pairids[,1], j = pairids[,2],
+                        betaij = rep(0,nrow(pairids)))
> for (i in 1:nrow(pairids)){
+   betaijs = (y[pairids[i,1]] - y[pairids[i,2]]) /
+     (x[pairids[i,1]] - x[pairids[i,2]])
+   pairbetas[i,3] = betaijs
+ }
> pairbeta_rm = pairbetas %>% filter(betaij != Inf)
> (betaTS = median(pairbeta_rm$betaij))
[1] 2.113208
> (alphaTS = median(y - betaTS * x))
[1] 4.55934
```

Computing the pairwise slopes for all $\binom{13}{2} = 78$ pairs of countries shows one pair with equal GDP (X) values. Excluding that pair, the median of the slopes gives $\hat{\beta}^{TS} = 2.11$, $\hat{\alpha}^{TS} = 4.55$: a much steeper slope than least squares (Figure 10.6 of the book compares the two lines). For reference, `mblm()` of the `mblm` package, `zyp.sen()` of `zyp` and `tsreg()` of `WRS` also compute the Theil–Sen estimates in simple regression:

```
> mblm::mblm(y ~ x, repeated = FALSE)

Call:
mblm::mblm(formula = y ~ x, repeated = FALSE)

Coefficients:
(Intercept)            x
      4.559        2.113
```

Finally the 95% interval for β from (10.46): compute the orders l, u from Kendall's tau distribution, then $(\beta_{(l)}, \beta_{(u)})$.

```
> (l = (1 + SuppDists::qKendall(0.025,n)) * choose(n,2)/2)
[1] 23
> (u = (1 + SuppDists::qKendall(1-0.025,n)) * choose(n,2)/2)
[1] 55
> (TS.CI = sort(pairbeta_rm$betaij)[c(l,u)])
[1] 0.3395639 4.1333333
```

Consistent with the test by Kendall's tau (rejecting $\tau = 0$ or $\beta = 0$), the interval for β does not contain 0.

The Theil–Sen estimator is not easily defined in multiple regression; various different estimation methods have been proposed. One using backfitting is briefly introduced. Assume the multiple

regression model

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.47}$$

Backfitting estimates β_j assuming $\hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p$ are already given. Of course none is given at first, so backfitting needs initial values; given them, β_j is estimated sequentially for $j = 0, 1, \ldots, p$, repeatedly.

1. For $j = 1, \ldots, p$ compute the Theil–Sen estimate $\hat\beta_j^{(0)}$ in the simple regression $y_i = \beta_0 + \beta_j x_{ij} + \epsilon_i$ (superscript "(0)" = initial value). The intercept is set to $\hat\beta_0^{(0)} = \text{median}\{y_i - \sum_{j=1}^p \hat\beta_j^{(0)} x_{ij}\}$.
2. For $k = 1, 2, 3, \ldots$, for each $j = 1, \ldots, p$, with response $z_i^{(k)} := y_i - \sum_{l \neq j} \beta_l^{(k-1)} x_{il}$ and explanatory variable x_{ij}, compute the Theil–Sen estimate in the simple regression $z_i^{(k)} = \beta_0 + \beta_j x_{ij} + \epsilon_i$ as $\hat\beta_j^{(k)}$. Finally $\hat\beta_0^{(k)} = \text{median}\{y_i - \sum_{j=1}^p \beta_j^{(k)} x_{ij}\}$.
3. Repeat step 2 until $\hat\beta_j^{(k)} - \hat\beta_j^{(k-1)}$ is very small for all j.

Example 10.19

The Theil–Sen estimator in multiple regression uses the backfitting algorithm above. Here we obtain the Theil–Sen estimates for the multiple regression explaining happiness of the 153 countries of Example 10.7 by 6 explanatory variables. The backfitting algorithm is implemented in `tsreg()` of the R package `WRS`.[a]

```
> h = read.csv("data/worldbank2020.csv")
> dat = h %>% dplyr::select(3:9)
> colnames(dat) = c("Score", "GDP", "Social",
+                   "Life", "Freedom", "Generosity", "Corruption
    ")
>
> ts.out = WRS::tsreg(as.matrix(dat[,2:7]), dat$Score,iter =
    100)
> ts.out$coef
[1] 1.5819695 0.7203707 1.4465492 1.0861433 1.0267905 1.2907304
    1.1652263
```

(`iter = 30` would force 30 iterations of backfitting.) Compared with least squares:

```
> lm(Score ~ ., data = dat)$coefficients
(Intercept)         GDP      Social        Life     Freedom
  1.8870774   0.7380488   1.1529984   0.9821926   1.4832835
 Generosity  Corruption
  0.6202784   0.9719126
```

Not a big difference, but most coefficient estimates went up a little. In the scatter plot matrix of Figure 10.3 several observations look like outliers; the Theil–Sen estimator removed their influence.

[a] The `WRS` package is not on CRAN; install it directly from https://github.com/nicebread/WRS.

Inference on coefficients with the Theil–Sen estimator in multiple regression has no particularly good method, but resampling (bootstrap or shuffling) can be used. The bootstrap, however, artificially produces many tied observations, so the accuracy of the bootstrap sampling distribution is not guaranteed. Permutation tests can be done without this tie problem.

Least trimmed squares estimators. The least squares estimator of the coefficients in the multiple regression model (10.47) is

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2 = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (e_i(\boldsymbol{\beta}))^2. \tag{10.48}$$

Like removing large values at both ends in a trimmed mean, removing the observations with large squared errors $e_i^2(\boldsymbol{\beta})$ in regression and taking the $\boldsymbol{\beta}$ with the smallest sum of squared errors gives the least trimmed squares (LTS) estimator. More precisely, with $e_{(i)}^2(\boldsymbol{\beta})$ the order statistics of the squared errors for each $\boldsymbol{\beta}$, the 10%-LTS estimator is

$$\hat{\boldsymbol{\beta}}^{LTS} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{h} e_{(i)}^2(\boldsymbol{\beta}), \qquad h = \lfloor n - 0.1n \rfloor. \tag{10.49}$$

The LTS estimator with $h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ trimmed observations is often recommended (Wilcox, 2011).

Several other robust estimators similar to LTS using squared errors have been proposed, e.g. the least median of squares (LMS) estimator minimizing the median of the squared errors. These have high breakdown points but low efficiency and are not used often.

Huber's M-estimator in regression. Replacing the "square" of the errors in the least squares estimator (10.48) of the multiple regression model by Huber's Ψ,

$$\hat{\boldsymbol{\beta}}^{\Psi} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \Psi\Big(\frac{y_i - \mathbf{x}_i'\boldsymbol{\beta}}{\hat{\tau}(\boldsymbol{\beta})}\Big), \tag{10.50}$$

is Huber's M-estimator in regression, where the error standard deviation estimate $\hat{\tau}(\boldsymbol{\beta})$ is

$$\hat{\tau}(\boldsymbol{\beta}) = \text{median}\{|y_i - \mathbf{x}_i'\boldsymbol{\beta}|\} \times 1.4826, \tag{10.51}$$

the MAD assuming the median of the $\{y_i - \mathbf{x}_i'\boldsymbol{\beta}\}$ is 0.

Huber's M-estimator is robust and highly efficient. Its relative efficiency to least squares (under normality) depends on the boundary value k of Ψ; at $k = 1.28$ it is known to be about 90%. The robustness of this M-estimator admits another interpretation: Huber's M-estimator is robust when the response has outliers but not when the explanatory variables do (high-leverage points). To see this, simply let $p = 1$, $\hat{\tau} = 1$; the derivative of $\Psi(y_i - \alpha - \beta x_i)$ in β is

$$\frac{d}{d\beta}\Psi(y_i - \alpha - \beta x_i) = -x_i \psi(y_i - \alpha - \beta x_i). \tag{10.52}$$

Even if y_i goes to infinity this derivative stays bounded, but as x_i grows it diverges. Because $|\psi(x)|$ is bounded, robustness holds, but not with respect to x_i. Yohai's MM-estimator, next, uses Tukey's biweight as $\psi(x)$ to be robust in the explanatory variables too.

Yohai's MM-estimator. Yohai (1987) devised an estimator keeping the merit of M-estimators (high efficiency) while robust to any outliers in explanatory and response variables. First, instead of Huber's function with $\psi(x) = \pm k$ for large x, use Tukey's biweight as $\psi(x)$. Tukey's biweight increases through 0 and then decreases again (Figure 10.4). Such functions are called redescending. In (10.52), as x_i grows the error $|y_i - \alpha - \beta x_i|$ grows and $\psi(y_i - \alpha - \beta x_i)$ becomes 0, so an observation with outlying x_i is removed. Yohai's MM-estimator in multiple regression is the solution of the system

$$\sum_{i=1}^{n} \psi^{\text{biweight}}\Big(\frac{y_i - \mathbf{x}_i'\boldsymbol{\beta}}{\hat{\tau}}\Big) x_{ij} = 0, \qquad j = 0, \ldots, p, \tag{10.53}$$

where $x_{i0} = 1$ and $\hat{\tau}$ is fixed as one of the robust estimators of the error standard deviation. Yohai allowed any high-breakdown estimator as $\hat{\tau}$: the MAD as in Huber's M-estimator, or another robust estimator such as the scale M-estimator or the S-estimator.[9] The MM-estimator also has no analytic solution and must be computed numerically; R's `MASS::rlm()` implements it with the S-estimator as $\hat{\tau}$. Tukey's biweight also needs a boundary value c; $c = 4.685$ is known to give 95% relative efficiency under normal errors, so this value is used (Yu and Yao, 2017). Yohai's MM-estimator, with breakdown point 50%, is one of the most robust estimators.

Huber's M-estimator and Yohai's MM-estimator both apply in multiple regression, but to see their difference clearly we use a simple regression example.

Example 10.20

For the simple regression of happiness on GDP of 13 countries, Huber's M-estimator and Yohai's MM-estimator via `rlm()` of `MASS`:

```
> wh = read.csv("data/whappiness.csv")
> x = wh$GDP.per.capita; y = wh$Score
> MASS::rlm(y ~ x, method = "M")$coefficients
(Intercept)           x
  6.4250015   0.6929371
> MASS::rlm(y ~ x, method = "MM")$coefficients
(Intercept)           x
  6.3118128   0.7783051
```

Compared with Theil–Sen, the slope estimates differ little from least squares: the observation judged an "outlier" is not an outlier from the M-estimator's viewpoint, nor far from the others for the MM-estimator. To see the difference more clearly, adjust that outlier's GDP to a very large 100 and re-estimate:

```
> x[9] = 100
> MASS::rlm(y ~ x, method = "M")$coefficients
 (Intercept)            x
 7.360972591 -0.001222253
> MASS::rlm(y ~ x, method = "MM")$coefficients
(Intercept)           x
   4.649281    2.056968
```

Figure 10.7 of the book compares the fitted lines in both cases with the least squares line.

10.4.4 Inference on coefficients with M-estimators

Standard errors of robust estimators in regression can be expressed only asymptotically, and are valid only for very large samples. But M- or MM-estimates of coefficients are computed by minimizing a weighted sum of squares, so the standard error approximation obtained as a byproduct of the weighted least squares computation can be used as the standard error estimate. The next example compares this approximation with the bootstrap estimate.

Example 10.21

Multiple regression with the data of Example 10.19. First the standard error estimates approximated by weighted least squares:

```
> MASS::rlm(Score ~ ., data = dat, method = "MM") %>% summary()

Call: rlm(formula = Score ~ ., data = dat, method = "MM")
```

[9] The discussion would be long, so robust scale estimators and S-estimators are not treated.

```
Residuals:
     Min       1Q   Median       3Q      Max
-1.81988 -0.33616  0.02929  0.32349  1.36173

Coefficients:
            Value  Std. Error t value
(Intercept) 1.9314 0.2221     8.6961
GDP         0.7363 0.2589     2.8441
Social      1.0480 0.2737     3.8296
Life        0.9553 0.3523     2.7115
Freedom     1.7037 0.4058     4.1979
Generosity  0.5601 0.4980     1.1246
Corruption  1.3417 0.4768     2.8138
```

The standard errors are in the `Std. Error` column. To bootstrap them, $B = 1999$ resamples:

```
> n = nrow(dat)
> set.seed(1)
> B = 1999
> MM.boot = replicate(B, {
+   boot.id = sample(1:n,n,replace = TRUE)
+   dat.boot = dat[boot.id,]
+   MASS::rlm(Score ~ ., data = dat.boot, method = "MM")$
    coefficients}
+ ) %>% t()
> apply(MM.boot,2,sd)
(Intercept)         GDP      Social        Life     Freedom
  0.3819560   0.2476741   0.4911845   0.4050225   0.4693110
 Generosity  Corruption
  0.6501763   0.5714038
```

The bootstrap estimates are more accurate, but roughly similar to the weighted-least-squares approximations.

Both the M- and MM-estimators of coefficients have asymptotically computable standard error estimates, so in principle the approximate-pivot permutation tests and bootstrap-t intervals of Section 10.3 can be used.

Example 10.22 (Bootstrap-t intervals with Yohai's MM-estimator, case bootstrap)

With the MM-estimator $\hat{\beta}_i$ of the multiple regression on the data of Example 10.19, the 95% bootstrap-t interval for β_i is

$$(\hat{\beta}_i - \widehat{\text{se}}(\hat{\beta}_i)Q^*(1-\alpha/2),\ \hat{\beta}_i - \widehat{\text{se}}(\hat{\beta}_i)Q^*(\alpha/2)), \tag{10.54}$$

where Q^* is the quantile function of the bootstrap replicates $(\hat{\beta}_i^* - \hat{\beta}_i)/\widehat{\text{se}}(\hat{\beta}_i^*)$, computed for all $i = 0, \ldots, p$:

```
> MM.out = summary(rlm(Score ~ ., data = dat, method = "MM"))
    [[4]]
> beta.hat = MM.out[,1]
> beta.hat.se = MM.out[,2]
```

These are the MM-estimates $\hat{\beta}_i$ and standard errors $\widehat{\text{se}}(\hat{\beta}_i)$. The code for Q^*:

```
> B = 1999
> set.seed(1)
```

```
> # resampling t
> MM.boot = replicate(B, {
+   boot.id = sample(1:n,n,replace = TRUE)
+   dat.boot = dat[boot.id,]
+   MM.out.boot = summary(rlm(Score ~ ., data = dat.boot, method
    = "MM"))[[4]]
+   (MM.out.boot[,1] - beta.hat)/MM.out.boot[,2]}
+ ) %>% t()
> ci = matrix(NA, nrow = 7, ncol = 2); colnames(ci) = c("2.5%","
    97.5%")
> row.names(ci) = colnames(MM.boot)
> for (i in 1:7){
+   ci[i,] = beta.hat[i] - beta.hat.se[i] * quantile(MM.boot[,i
    ],probs=c(1-0.025,0.025))
+ }
> ci
                 2.5%    97.5%
(Intercept)  1.26065294 2.816903
GDP          0.25383080 1.249724
Social       0.00938614 1.975257
Life         0.07427750 1.761928
Freedom      0.68511127 2.634962
Generosity  -0.82880966 1.785123
Corruption   0.34732311 2.611980
```

Except for the coefficient β_5 of Generosity, all 95% intervals exclude 0.

Example 10.23 (Permutation tests of coefficients with Yohai's MM-estimator, shuffling observations)

With the MM-estimates of the multiple regression on the data of Example 10.19, random permutation tests at level 5% of

$$H_0 : \beta_i = 0 \quad \text{vs} \quad H_1 : \beta_i \neq 0 \tag{10.55}$$

for each coefficient β_i $(i = 0, \ldots, p)$. The code shuffles the observations to approximate the null distribution of $T_i = (\hat{\beta}_i - 0)/\widehat{\text{se}}(\hat{\beta}_i)$ and computes the p-values:

```
> (t.stats = beta.hat / beta.hat.se)
(Intercept)         GDP      Social        Life     Freedom
   8.696273    2.844139    3.829285    2.711510    4.198152
 Generosity  Corruption
   1.124606    2.814131
> set.seed(1)
> M = 1999
> MM.perm = replicate(M,{
+   perm.id = sample(1:n,n, replace = FALSE)
+   dat.perm = dat
+   dat.perm$Score = dat$Score[perm.id]
+   MM.out.perm = summary(rlm(Score ~ ., data = dat.perm, method
    = "MM"))[[4]]
+   (MM.out.perm[,1])/MM.out.perm[,2]}
+ ) %>% t()
> perm.pvalues = vector(length = 7)
> for (i in 1:7){
+   perm.pvalues[i] = mean( abs(c(t.stats[i],MM.perm[,i])) >=
    abs(t.stats[i]))
+ }
```

```
> perm.pvalues
[1] 1.0000 0.0070 0.0010 0.0110 0.0005 0.2480 0.0060
```

Only the hypothesis on β_5 (Generosity) has $p = 0.248$ and is not rejected; all other coefficients differ significantly from 0 at 5%. This agrees with the conclusion of Example 10.22. (The intercept's permutation p-value is 1 because shuffling the response does not alter the intercept test's null behavior in a useful way: shuffling the observations tests $\beta_1 = \cdots = \beta_p = 0$ jointly, not $\beta_0 = 0$, as noted in Section 10.3.1.)